# Saverio Mattia **Merenda**

MASTER'S STUDENT

*Reggio Emilia - Parma, Italy*

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## Education

**University of Parma** *Parma, Italy*

M.S. IN COMPUTER SCIENCE, EXPECTED GRADUATION YEAR: JULY 2026 *From Sep. 2024*

- Topics: **Software Verification**, **Machine Learning & AI**.
- Best Marks (so far): **Compilers**, **Quantum Computing**, **Big Data & Data Mining**, **AI**.

**University of Parma** *Parma, Italy*

B.S. IN COMPUTER SCIENCE, GRADE: 108/110 *Sep. 2021 - Jul. 2024*

- Thesis Title: *Construction of complete Control-Flow Graphs for bytecode EVM.*
- Best Marks: **Calculus**, **Programming**, **Algorithms & DS**, **Cloud Administration**, **Database**, **AI**, **HPC**.
- Got different ERGO scholarships given to promising Unipr students.

## Experience

**University of Parma** *Parma, Italy*

RESEARCHER AND SOFTWARE DEVELOPER *From Jan. 2025*

- **Development of xEVMLiSA:** Currently working on the early stages of xEVMLiSA, a project aimed at expanding EVMLiSA to analyze smart contracts across multiple blockchains (cross-chain).

**University of Parma** *Parma, Italy*

TEACHING ASSISTANT AND TUTOR *Sep. 2024 - Jul. 2025*

- Teaching Assistant and Tutor for the "Fondamenti di Programmazione A + B (15 CFU)" bachelor course in the Computer Science degree.

**University of Parma - Internship** *Parma, Italy*

RESEARCHER AND SOFTWARE DEVELOPER *Sep. 2023 - May 2025*

- **Development of EVMLiSA:** Contributed significantly to the development of EVMLiSA, a static analyzer based on LiSA (Library for Static Analysis) that uses abstract interpretation to analyze Ethereum smart contracts. Its main goal is to build sound Control-Flow Graphs (CFGs) to support the creation of checkers for detecting vulnerabilities (such as reentrancy, randomness dependency, ecc.). Several checkers have been designed and implemented to enhance the analysis of potential security issues.
- **In-depth study of Static Analysis and Abstract Interpretation:** Gained in-depth knowledge of static analysis and abstract interpretation, providing the theoretical foundation necessary for contributing to EVMLiSA.
- **Thoroughly study of design patterns for software development:** Learned and applied key design patterns and software frameworks to develop reliable and scalable applications, later integrating them into EVMLiSA to enhance its architecture and maintainability.

**HeliopsyLab** *Reggio Emilia, Italy*

SOFTWARE DEVELOPER *Jan. 2021 - Sep. 2021*

- Developed and implemented cutting-edge management software for hospital facilities in the Lazio and Lombardia regions, streamlining operations and reducing administrative inefficiencies by 15%.
- Advanced new features within the Picasso and Mirth integration software, enhancing its capabilities and functionality.

**Pico srl** *Reggio Emilia, Italy*

SOFTWARE DEVELOPER *May. 2019 - Jul. 2019*

- Engineered a user-friendly application for efficient management of diverse user profiles within a database. Enhanced security measures and optimized the existing system for improved performance.
- Designed an algorithm to construct a wizard-style exam, streamlining the assessment process for a more user-friendly experience.

## Pubblications

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| 2025 | **Blockchain: Research and Applications**, V. Arceri, S. M. Merenda, L. Negrini, L. Olivieri, E. Zaffanella: *"EVMLiSA: Sound Static Control-Flow Graph Construction for EVM Bytecode".* (doi: 10.1016/j.bcra.2025.100384) | *Journal paper* |
| 2024 | **FTfJP@ISSTA/ECOOP 2024**, V. Arceri, S. M. Merenda, G. Dolcetti, L. Negrini, L. Olivieri, E. Zaffanella: *"Towards a Sound Construction of EVM Bytecode Control-flow Graphs".* (doi: 10.1145/3678721.3686227) | *Workshop paper* |

## Talks

### International Conferances and Workshop

2024 **Towards a Sound Construction of EVM Bytecode Control-flow Graphs**, 26th International Workshop on Formal Techniques for Java-like Programs, FTfJP 2024. [slides] *Vienna, Austria*

## Research Projects

**Research Participant, ”LLMs Meet Static Analysis: improving quality and reliability of AI-generated code”** *Parma, Italy*

ISCRA Project (Class C), Cineca *2024*

The goal of the project is to conduct an extensive quality and safety evaluation of the code generated with some of the most popular and open-source LLMs employing static analyzers, that can detect vulnerabilities and run-time errors statically, without executing the code. Once this information is available, it will be included in the code-generation task, to guide the LLM itself to produce a more precise and safe output, in which static analysis is somehow introduced in the pipeline of the code-generation task.

## Tools and Software

**EVMLiSA: an abstract interpretation-based static analyzer for EVM bytecode**

Java, Gradle *From Sep. 2023*

EVMLiSA is a specific implementation of a static analyzer using the LiSA (Library for Static Analysis) library to conduct static analysis of Ethereum Virtual Machine (EVM) bytecode. In particular, it is dedicated to generate a sound Control-Flow Graph (CFG) of smart contracts deployed on the Ethereum blockchain. EVMLiSA's primary objective is to provide semantic information and valuable warnings for developers and security auditors. EVMLiSA is distributed under the MIT license, and it is available on GitHub: `github.com/lisa-analyzer/evm-lisa`.

## Events

2025 **Conference participant**, CSV 2025, 4rd Challenges of Software Verification Symposium 2025. [link] *Venice, Italy*

2024 **Student**, Lipari Summer School on Abstract Interpretation. [link] *Lipari, Italy*

2024 **Conference participant**, CSV 2024, 3rd Challenges of Software Verification Symposium 2024. [link] *Venice, Italy*

## Academic Activity

**Isolette: Interactive Scenario Visualization and UI Simulation**

C, Vue.js, Typescript *2025*

- Isolette is a model project for the development of safety-critical software, simulating a neonatal incubator system that monitors and controls temperature to ensure infant health. The project emphasizes formal specifications, deterministic behavior, and testability under strict requirements.
- Contributed to the User Interface Hardware (UIH) layer, defining and implementing both the UIH and its simulator (UIHS) to ensure behavioral consistency during simulation.
- Developed the web UI in Vue and TypeScript, with strict client-side JSON validation and a clear interface displaying time, messages, real-time temperature charts, and a time slider for interactive inspection.
- Designed and implemented a Scenario Visualization Framework to replace gnuplot, leveraging Vue, Chart.js, and modular design to produce dynamic, interactive, and navigable scenario visualizations.
- The result is a scalable, browser-based visualization frontend that improves usability, speeds up development, and removes external dependencies.

**Impact of Data Preprocessing on Neural Network Performance**

Python *2025*

- Investigated the effects of various data preprocessing techniques (NaN handling, outlier removal, normalization, transformation) on the performance of feed-forward neural networks.
- Implemented and compared multiple preprocessing scenarios on diverse classification and regression datasets, measuring impact using standard evaluation metrics.
- Key Concepts and Skills: Data Preprocessing, Neural Networks, Performance Evaluation, NaN Imputation, Outlier Removal (Isolation Forest), Normalization (Z-score), Quantile Transformation, Classification Metrics (Accuracy, Precision, Recall, F1), Regression Metrics (MAE, MSE, $R^2$).
- Source Code: `github.com/merendamattia/neural-network-performance-by-data-quality`

### GraphQL API Design and Comparative Analysis

GraphQL, REST — *2025*

- Designed and prototyped a GraphQL API, defining a schema using SDL and implementing queries/mutations for efficient, client-specified data retrieval.
- Analyzed and contrasted the GraphQL approach with traditional RESTful patterns, highlighting improvements in data fetching efficiency, reduced network calls, and prevention of over/under-fetching.
- Key Concepts and Skills: GraphQL, API Design, Schema Definition Language (SDL), Queries, Mutations, REST API Comparison, Data Fetching Optimization, Single Endpoint Architecture.

### Quantum Portfolio Optimization

Python — *2025*

- Explored quantum computing approaches to portfolio optimization, leveraging QAOA and VQE to solve Quadratic Unconstrained Binary Optimization (QUBO) problems.
- Designed and implemented simulations using Qiskit to evaluate algorithm performance under noiseless and noisy conditions.
- Highlighted quantum computing's potential for finance and the limitations imposed by current hardware.
- Source Code: `github.com/merendamattia/quantum-portfolio-optimization`

### Optimization of Academic Guarantors: a Declarative Approach

ASP, Python — *2024*

- Designed an automated system for assigning academic guarantors to university courses, adhering to complex ministerial regulations.
- Developed a robust data preprocessing pipeline and implemented the solution using Answer Set Programming (ASP) to dynamically optimize resource allocation.
- Ensured scalability across various datasets, reducing reliance on external contracted staff while meeting institutional and regulatory requirements.
- Source Code: `github.com/merendamattia/ottimizzazione-garanti-accademici`

### Deep Neural Network

C++, Python — *2024*

- The DNN (Deep Neural Network) project is designed to create a neural network framework in C++ and it provides tools to define, train, and evaluate neural network models.
- This project enables training neural networks using backpropagation and gradient descent algorithms. Users can specify network architectures, activation functions, and training parameters. Additionally, it offers features for loading and saving trained models.
- Source Code: `github.com/merendamattia/deep-neural-network`

## Extracurricular Activity

### My-gpt4

Python — *2023*

- Crafted a cutting-edge Python integration leveraging multiple reverse-engineered language-model APIs, contributing to the decentralization of the AI industry.
- Source Code: `github.com/merendamattia/my-gpt4`

### Tracking Messages on Bitcoin Blockchain using OP-RETURN field

JS, HTML — *2022*

- Devised an innovative algorithm to securely trace immutably written messages within the Bitcoin blockchain, utilizing transaction hashes for enhanced transparency.
- Demo: `merendamattia.com/dev/btc/tracking/`
- Source Code: `github.com/merendamattia/op-return-tracking-message-bitcoin`

### Development of Various Bots

Python, JS — *From 2021*

- Developed a sophisticated bot designed to navigate financial markets, employing strategic approaches to optimize profits while minimizing potential losses.
- Customized Telegram bots designed for personal optimization, streamlining and enhancing my daily routine.